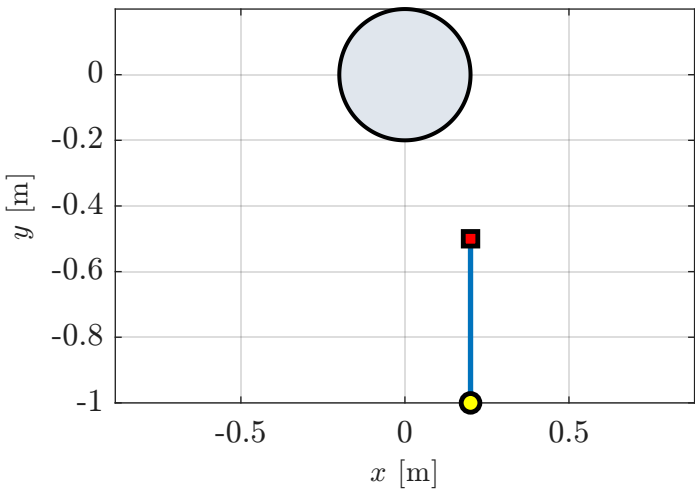
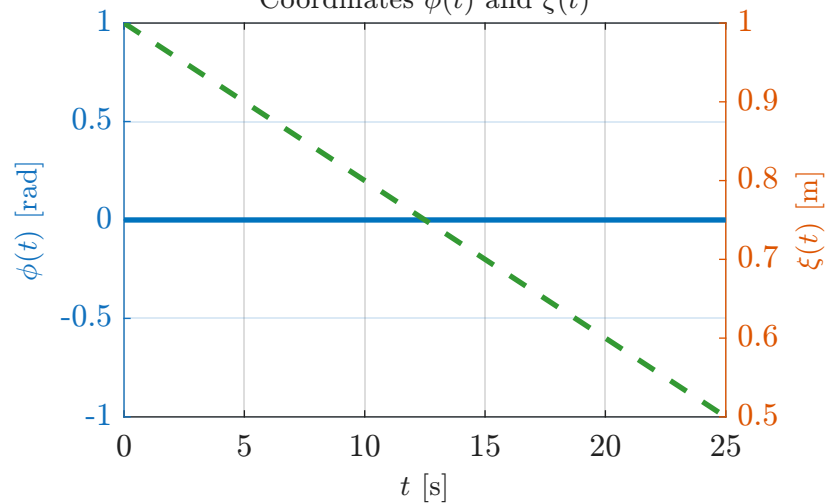


Case 1: Comprehensive Dynamics Summary ($\omega = 0.1$ rad/s, $\dot{\phi}_0 = 0$ rad/s)

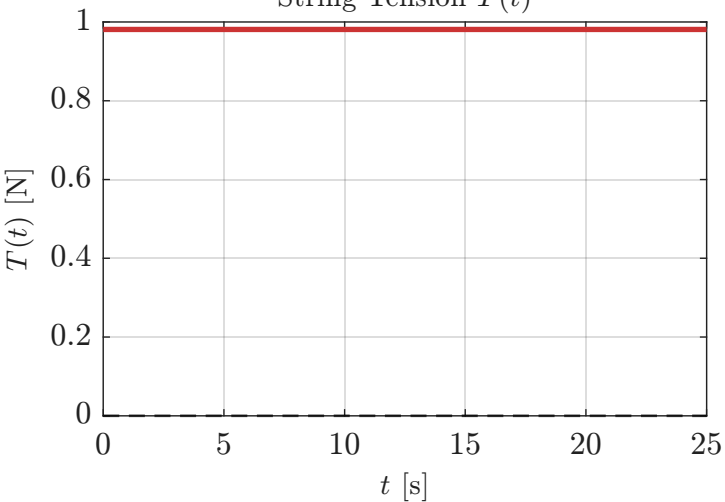
Trajectory in Oxy



Coordinates $\phi(t)$ and $\xi(t)$



String Tension $T(t)$



Mechanical Energy Breakdown

